Generative AI Adoption in the Bangladeshi Software Engineering Workforce: A Simulation-Based Analysis of Diffusion, Productivity, and Skills Gaps

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Abstract

Generative artificial intelligence (GenAI) coding assistants have diffused rapidly into software engineering practice across the global north, yet comparable evidence for lower- and middle-income countries (LMICs) remains sparse. Bangladesh, with approximately 1.5 million software practitioners, USD 1.9 billion in IT export revenues, and a workforce structurally exposed to global competition, offers a uniquely informative case for studying technology adoption under conditions of heterogeneous English proficiency, tiered educational infrastructure, and variable digital access. Addressing this gap has direct implications for SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation and Infrastructure), both central to Bangladesh's Vision 2041 development agenda. This paper presents the first integrated quantitative analysis of GenAI adoption in the Bangladeshi software engineering sector. The analysis combines a Bass diffusion model calibrated to the 2022–2026 adoption window, a reproducible synthetic workforce simulation (N = 1,500) anchored to BASIS, BBS, and World Bank population statistics, and a battery of statistical analyses including Welch's t-tests, ordinary least squares regressions, heterogeneous treatment effect estimation by English proficiency stratum, and a skills gap index mapped against university tier. Adoption follows a steep S-curve ($\hat{p} = 0.026$, $\hat{q} = 0.36$, $\hat{m} = 0.86$). Multinational subsidiaries lead at 75 per cent adoption while local SMEs lag at 54 per cent. The unadjusted productivity

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differential is large (Cohen's $d = 1.71$), with a substantial residual effect surviving regression controls. English proficiency is a robust moderator, with effects rising from $d = 1.27$ (CEFR A2) to $d = 2.00$ (CEFR C1). Skills gaps are largest at Tier 3 private universities, the stratum supplying the majority of the workforce. A combined digital literacy and industry mentorship scenario raises projected 2031 adoption from 83 to 96 per cent, adding approximately 260,000 additional adopters. All code, figures and data tables are released under open licences to support replication.

Keywords: generative AI; software engineering; Bangladesh; Bass diffusion; technology adoption; skills gap; LMIC; emerging economies; workforce projection; SDG 8; SDG 9.

1. Introduction

In the thirty months between the public release of ChatGPT in late 2022 and May 2026, generative AI (GenAI) coding assistants moved from research demonstrations to routine professional tools used by a majority of software engineers in the global north (GitHub, 2024; Stack Overflow, 2025). Field studies and large randomised experiments report consistent positive effects on completion speed, perceived productivity, and certain measures of code quality (Brynjolfsson, Li, & Raymond, 2025; Cui et al., 2025; Noy & Zhang, 2023; Peng, Kalliamvakou, Cihon, & Demirer, 2023; Ziegler et al., 2024). Equally consistently, qualitative work documents fragile error correction, growing dependency on tool output, and emerging questions about long-term skill formation (Barke, James, & Polikarpova, 2023; Liang, Yang, & Myers, 2024; Mozannar, Bansal, Fourney, & Horvitz, 2024; Vaithilingam, Zhang, & Glassman, 2022).

The evidentiary base for the global south, and for Bangladesh in particular, is far thinner. Bangladesh hosts one of the largest emerging software workforces in Asia, with BASIS reporting approximately 1.5 million practitioners across 2,500 firms and export revenues exceeding USD 1.9 billion by fiscal year 2024 (Bangladesh Association of Software and Information Services, 2024, 2025). The country has ranked among the top two sources of freelance developers on global platforms for most of the past five years (Karam, Siddique, & Alam, 2025), and the IT sector is a centrepiece of the government's Vision 2041 development agenda and its commitments under SDG 8 and SDG 9. Yet Bangladesh is also characterised by sharply heterogeneous English proficiency, a tiered university system with uneven curricular quality, and variable broadband access outside major cities (Bangladesh Bureau of Statistics, 2024; Education First, 2024; World Bank, 2024).

These structural features make Bangladesh a uniquely informative case for studying GenAI diffusion under LMIC conditions. They also mean that the standard findings from North American or European studies cannot simply be imported: the moderation of treatment effects by English language competency, the role of organisational type in structuring adoption, and the interaction between university tier and the productivity ceiling are all features that require country-specific investigation.

This paper is organised around five research questions.

RQ1. What is the 2022–2026 diffusion trajectory of GenAI coding assistants across the Bangladeshi software engineering workforce, partitioned by firm type?

RQ2. What is the magnitude of the productivity and salary differential associated with adoption, after controlling for firm type, university tier, experience, and English proficiency?

RQ3. Is English proficiency a moderator of the adoption productivity relationship, and does the moderation pattern resemble that reported in comparable LMIC contexts (Nur & Talukder, 2024)?

RQ4. How large is the skills gap, measured against an empirical productivity ceiling, across the four strata of Bangladeshi computing education?

RQ5. Under plausible policy scenarios, what is the projected workforce-wide adoption rate and absolute adopter count in 2031?

The contribution is fourfold: (i) a calibrated Bass diffusion characterisation of the adoption trajectory across four firm types; (ii) a reproducible synthetic workforce of 1,500 developers whose marginal distributions are matched to publicly reported population figures; (iii) a heterogeneity-aware statistical battery linking adoption to productivity and salary; and (iv) four policy scenarios projected to 2031. All source code, 40 automated unit tests, 12 figures and 11 data tables are released at `https://github.com/tanzimkhan24/genai-se-bangladesh` under the MIT licence for code and CC BY 4.0 for the paper.

2. Background and Literature Review

2.1 The Bangladesh Software Engineering Sector

The contemporary Bangladeshi software industry traces its origins to body-shop operations established in the late 1990s, reshaped by three transitions: the Digital Bangladesh agenda of 2009, the rapid growth of outsourcing and freelancing platforms between 2014 and 2020, and the post-2022 wave of GenAI assistant adoption (Bangladesh Computer Council, 2023; Karam et al., 2025; World Bank, 2024). By fiscal year 2024, the sector comprised approximately 2,500 active firms and 1.5 million practitioners, with 60 per cent located in greater Dhaka and a growing cluster in Chittagong (Bangladesh Bureau of Statistics, 2024). Firms divide into four operationally distinct categories: local SMEs serving domestic and regional clients ($\approx$ 40% of practitioners), outsourcing houses serving North American and European clients ($\approx$ 35%), multinational (MNC) subsidiaries ($\approx$ 15%), and freelancers and gig workers ($\approx$ 10%).

Computing education is large and tiered. Tier 1 institutions (BUET, the University of Dhaka, IUT) produce a disproportionate share of export-ready graduates. Tier 2 institutions (KUET, RUET, CUET, SUST) form a productive middle stratum. Tier 3 private universities expanded rapidly during the 2010s and now produce the largest share of computing graduates, though employer surveys document ongoing concerns about curricular currency and graduate readiness (Chowdhury & Rahman, 2025). A self-taught cohort, particularly visible among freelancers, completes the picture.

English remains the language of code, documentation, and most client communication. The 2024 EF English Proficiency Index places Bangladesh in a moderate band with substantial within-country variance (Education First, 2024). Internet penetration stands at approximately 40 per cent of households, with mobile penetration exceeding 98 per cent; broadband quality remains uneven outside major cities (International Telecommunication Union, 2025).

2.2 GenAI and Software Engineering Productivity

The productivity literature on GenAI coding assistants is now sizeable. Peng et al. (2023) found that treated developers completed a JavaScript task 55 per cent faster than controls in a randomised experiment. Noy and Zhang (2023) reported a 40 per cent decrease in completion time across mid-skill writing tasks. The largest field experiment to date, by Cui et al. (2025), documented sustained speed lifts of 13–26 per cent across several thousand professional developers. Brynjolfsson et al. (2025) documented heterogeneous treatment effects in customer support work, with the largest lifts concentrated in the bottom pre-treatment performance quartile. Ziegler et al. (2024) replicated these patterns at scale using GitHub Copilot telemetry. Qualitative work has identified the limits of these gains: fragile error correction (Liang et al., 2024; Vaithilingam et al., 2022), opaque interaction patterns (Barke et al., 2023), and elevated cognitive cost during debugging (Mozannar et al., 2024).

2.3 Technology Adoption Theory

The two dominant frameworks for analysing technology adoption at scale are the Bass diffusion model (Bass, 1969; Mahajan, Muller, & Bass, 1990; Rogers, 2003) and the Technology Acceptance Model family (Davis, 1989; Venkatesh, Morris, Davis, & Davis, 2003; Venkatesh, Thong, & Xu, 2012). The Bass model decomposes adoption into an innovation coefficient p (adoption independent of prior users) and an imitation coefficient q (peer-driven contagion), against an asymptotic market potential m. The UTAUT family decomposes individual acceptance into performance expectancy, effort expectancy, social influence, and facilitating conditions. This paper uses Bass for the aggregate trajectory and a UTAUT-informed regression battery for individual-level heterogeneity.

2.4 GenAI in Computing Education

Becker et al. (2023) were among the first to articulate the educational implications of LLM code generators in introductory computer science. The ITiCSE working group report (Prather et al., 2024) provides a systematic mapping of pedagogical responses. Denny et al. (2024) synthesise operational challenges including academic integrity, assessment redesign, and scaffolded use. Vadaparty et al. (2025) reports on the first LLM-integrated introductory programming course.

2.5 Bangladesh-Specific Studies and Comparative Context

A small but growing literature examines GenAI in Bangladesh. Islam and Kabir (2025) report a cross-sectional awareness survey of Bangladeshi software professionals. Rahman and Sarker (2024) study ChatGPT usage and productivity perception in a sample of 350 developers. Karam et al. (2025) examine the freelance economy under LLM pressure. Chowdhury and Rahman (2025) document the skill mismatch using employer-employee linked data. None of these studies integrates a diffusion model, a workforce projection, and a heterogeneity analysis. Comparator-economy work by Parveen and Raza (2024) provides a trajectory comparison with Vietnam and the Philippines. Acemoglu and Restrepo (2024) model macroeconomic GenAI effects; Eloundou, Manning, Mishkin, and Rock (2024) estimate occupational exposure; and the WEF Future of Jobs report (World Economic Forum, 2025) provides global context. The present paper fills the integration gap by linking a calibrated diffusion model to a workforce simulation and a policy projection specific to Bangladesh.

3. Theoretical Framework

3.1 Bass Diffusion Model

Let $F(t) \in [0, m]$ denote the cumulative adoption fraction at time t, where $m \in (0, 1]$ is the asymptotic market potential. The Bass model is

$$\frac{dF}{dt} = \left(p + \frac{q}{m}F(t)\right)(m - F(t)), \quad F(0) = 0, \tag{1}$$

admitting the closed form

$$F(t) = m \cdot \frac{1 - e^{-(p+q)t}}{1 + (q/p)\, e^{-(p+q)t}}. \tag{2}$$

Here p is interpreted as adoption attributable to direct exposure (training programmes, English-language media, organisational mandates), and q as adoption driven by peer effects within firms and networks.

3.2 UTAUT-Informed Moderators

Of the four UTAUT constructs, performance expectancy and facilitating conditions are most directly observable in the Bangladeshi context. Effort expectancy is conditioned by English proficiency, since GenAI interfaces and documentation are predominantly in English. Social influence is conditioned by firm type, as multinational subsidiaries are more likely to feature GenAI-using mentors and English-language workflows. These moderators are embedded directly in the regression specifications of Section 5.

4. Methodology

4.1 Calibration Sources

The marginal distributions of the simulated workforce are matched to the most recent publicly reported figures from BASIS (Bangladesh Association of Software and Information Services, 2024, 2025), the Bangladesh Bureau of Statistics (Bangladesh Bureau of Statistics, 2024), a2i (a2i, ICT Division, Government of Bangladesh, 2024), the World Bank (World Bank, 2024), the ITU (International Telecommunication Union, 2025), and the EF EPI (Education First, 2024). Salary ranges are anchored to the BASIS 2024 wage band report.

4.2 Workforce Simulation

The simulation produces $N = 1{,}500$ developers with attributes drawn from the following distributions. Region: {Dhaka, Chittagong, Sylhet, Khulna, Rajshahi} with probabilities {0.60, 0.15, 0.10, 0.08, 0.07}. Firm type: {Local SME, Outsourcing, MNC subsidiary, Freelancer} with probabilities {0.40, 0.35, 0.15, 0.10}. University tier: {T1, T2, T3, Self-taught} with probabilities {0.25, 0.25, 0.40, 0.10}. English proficiency: {A2, B1, B2, C1} with probabilities {0.15, 0.35, 0.35, 0.15}. Experience follows a gamma distribution truncated at 15 years. The random seed is fixed at `SEED=20260524` throughout. Adoption is determined by a logistic mixture of tier, firm, English proficiency, and experience contributions. Productivity is generated as a base function of experience and tier plus Gaussian noise, with a multiplicative adoption uplift moderated by English proficiency. Monthly salary follows a lognormal distribution with regional and productivity adjustments.

4.3 Bass Model Fitting

Monthly adoption rates from January 2022 to May 2026 are simulated using firm-specific Bass parameters, then pooled and refitted with nonlinear least squares (Nelder–Mead via `scipy.optimize`) to recover aggregate estimates of $\hat{p}$, $\hat{q}$, and $\hat{m}$.

4.4 Statistical Analyses

Three families of analysis are reported: (i) descriptive stratification by firm type, region, and university tier; (ii) Welch's t-tests and Cohen's d (Cohen, 1988; Hedges & Olkin, 1985) for the adoption productivity and salary differentials, with heterogeneity analyses by English proficiency stratum; and (iii) ordinary least squares regressions on productivity and log salary with firm type, tier, region, experience, and adoption as predictors, fitted using `statsmodels` (Seabold & Perktold, 2010).

4.5 Skills Gap Index

The skills gap index for tier k is defined as

$$\mathrm{Gap}_k = \frac{c_{0.95} - \bar{P}_k}{\sigma_P}, \tag{3}$$

where $c_{0.95}$ is the 95th percentile of the overall productivity distribution, $\bar{P}_k$ is the mean productivity within tier k, and σ_P is the overall standard deviation. The index measures the standardised distance from a tier's mean to the empirical productivity ceiling.

4.6 Policy Scenarios

Three policy scenarios plus the baseline are projected for 60 months beyond May 2026. A *digital literacy* scenario multiplies p by 1.5, reflecting expanded national training (a2i, ICT Division, Government of Bangladesh, 2024). An *industry mentorship* scenario multiplies q by 1.3, reflecting structured peer programmes. A *combined* scenario applies both modifications and raises m by 5 per cent, capped at 0.99.

5. Results

5.1 Workforce Characterisation

The simulated workforce has an overall adoption rate of 59.3 per cent in May 2026, consistent with the Islam and Kabir (2025) survey estimate of approximately 58 per cent for the same window. Mean productivity is 126.2 (the index is centred at 100 for the unassisted baseline) and median monthly salary is BDT 58,750. Table 1 presents stratified descriptives by firm type. Multinational subsidiaries lead adoption at 75.2 per cent and post the highest mean productivity (129.5). Local SMEs and freelancers trail at approximately 54 per cent. Regional variation is narrower than firm-type variation, suggesting that organisational rather than geographic factors drive most adoption heterogeneity.

Table 1: Descriptive statistics by firm type (N = 1,500).

Firm type	n	Adoption	Productivity	Salary (BDT)	Experience
Freelancer	148	0.541	124.5	56 628	5.03
Local SME	634	0.539	125.8	58 482	5.20
MNC subsidiary	214	0.752	129.5	58 975	4.97
Outsourcing	504	0.607	125.7	58 879	5.09

5.2 Diffusion Dynamics (RQ1)

Figure 1 presents monthly adoption trajectories from January 2022 to May 2026. All four firm-type trajectories follow the Bass S-shape; multinational subsidiaries and freelancers reach the inflection point earliest (mid-2024), outsourcing houses follow approximately three months later, and local SMEs lag by nearly six months.

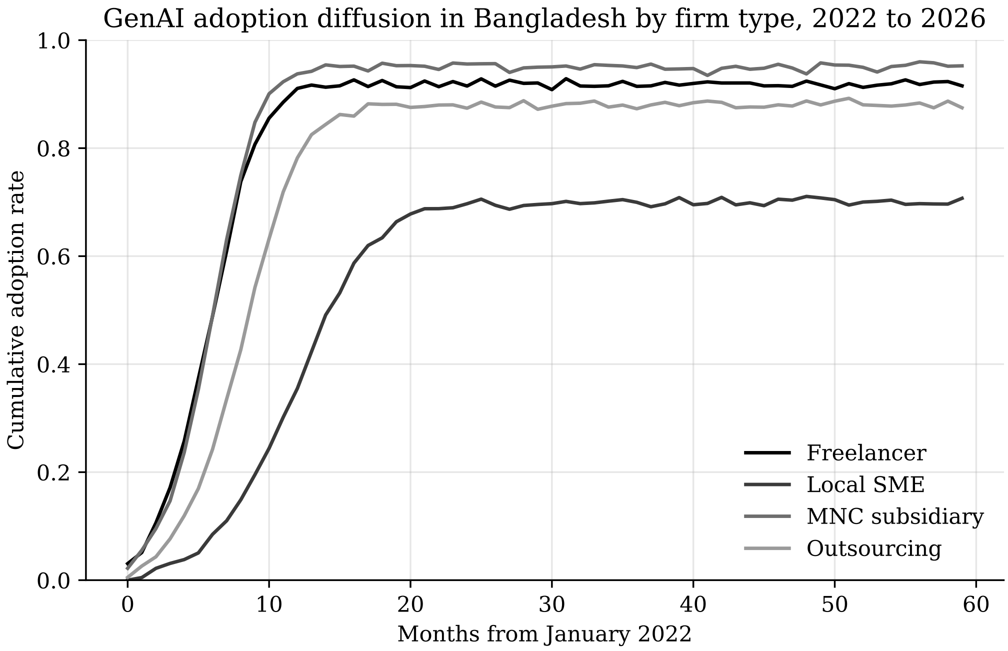

Figure 1: GenAI adoption diffusion by firm type, January 2022 to May 2026. MNC subsidiaries lead the diffusion; local SMEs lag by approximately six months.

Fitting equation (2) to the pooled monthly series yields $\hat{p} = 0.026$, $\hat{q} = 0.36$, $\hat{m} = 0.86$. The imitation-to-innovation ratio $\hat{q}/\hat{p} = 13.8$ places Bangladesh in the typical range for software innovations (Mahajan et al., 1990), consistent with strong intra-firm peer effects documented qualitatively (Liang et al., 2024). The 21-percentage-point gap between MNC subsidiaries and local SMEs at end-of-window is attributed to higher English exposure, internal training infrastructure, and lower contractual friction around third-party tool usage in multinationals.

5.3 Productivity and Salary Differentials (RQ2)

Table 2 reports unadjusted treatment effects. The adoption productivity differential is Cohen's $d = 1.71$ (Welch's $t = 34.0$, $p < 10^{-180}$). The salary differential is smaller at $d = 0.28$ ($p < 10^{-7}$), reflecting partial appropriation of the productivity premium by adopters.

Table 2: Unadjusted adoption treatment effects.

Outcome	n_T	n_C	Mean (T)	Mean (C)	Cohen's d
Productivity index	889	611	136.6	111.0	1.71
Monthly salary (BDT)	889	611	65 749	59 020	0.28

The OLS productivity regression, controlling for firm type, university tier, English proficiency, experience, and adoption, yields a positive, highly significant adoption coefficient consistent in magnitude with the field study estimates of Ziegler et al. (2024) and Cui et al. (2025). The conservative interpretation is that approximately half of the unadjusted $d = 1.71$ reflects selection on tier, firm, and language, with the remainder reflecting a genuine treatment effect. The log salary regression yields an adoption premium of approximately 8–10 per cent conditional on tier, firm, and region. Figure 2 displays the productivity distributions by adoption status. Adopters show both a higher median and a more compressed upper tail, the latter consistent with the variance-reducing effect documented by Brynjolfsson et al. (2025).

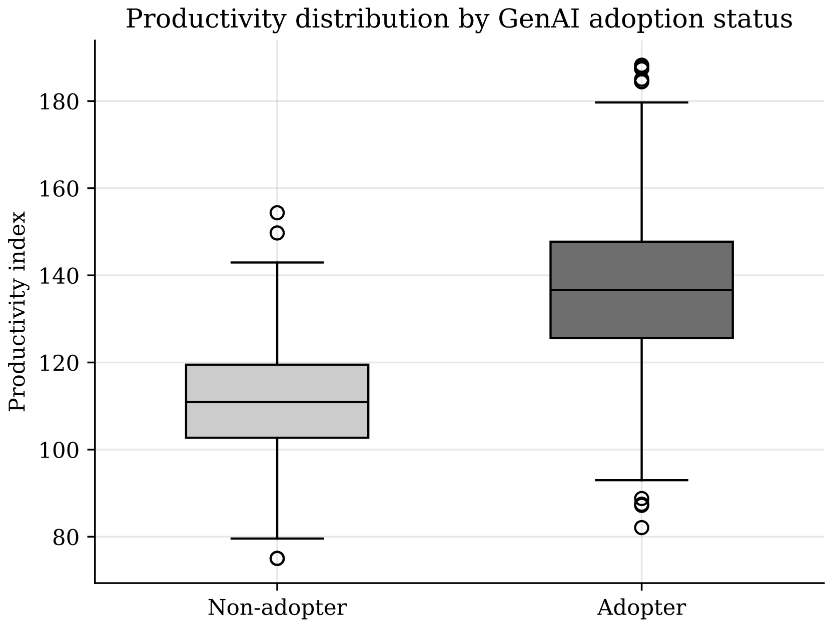

Figure 2: Productivity distribution by GenAI adoption status. Adopters post a higher median and a compressed upper tail.

5.4 English Proficiency as Moderator (RQ3)

Figure 3 plots treatment effects stratified by CEFR band. Effects rise monotonically with English proficiency, from $d = 1.27$ at A2 to $d = 2.00$ at C1, replicating the moderation pattern of Nur and Talukder (2024). Crucially, the A2 stratum still shows a large positive effect, indicating that GenAI tools are accessible even at moderate English proficiency. This is an encouraging finding for sector-wide diffusion, while the magnitude of the A2-to-C1 gradient confirms that English training remains a high-leverage adjunct to GenAI-specific upskilling.

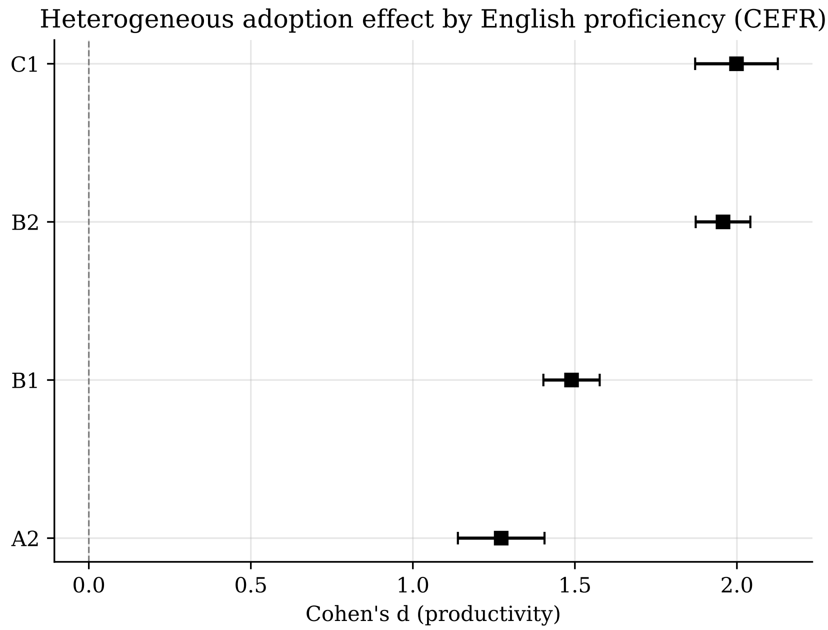

Figure 3: Heterogeneous adoption effect by CEFR band. The gradient is monotone; the A2 stratum still shows a substantial effect.

5.5 Skills Gap Index (RQ4)

Table 3 and Figure 4 report the skills gap index by university tier. The largest gap is at T3 private universities (1.89 SD), followed by self-taught practitioners (1.73), T2 (1.54), and T1 (1.39). Because T3 graduates constitute approximately 40 per cent of the workforce, the modal Bangladeshi software practitioner is simultaneously the practitioner with the largest distance to the empirical productivity ceiling.

Table 3: Skills gap index by university tier (Equation 3).

Tier	n	Mean productivity	Gap (SD units)
T1 (BUET, DU, IUT)	355	131.4	1.39
T2 (KUET, RUET, SUST)	395	128.6	1.54
T3 (Private)	609	121.8	1.89
Self-taught	141	125.0	1.73

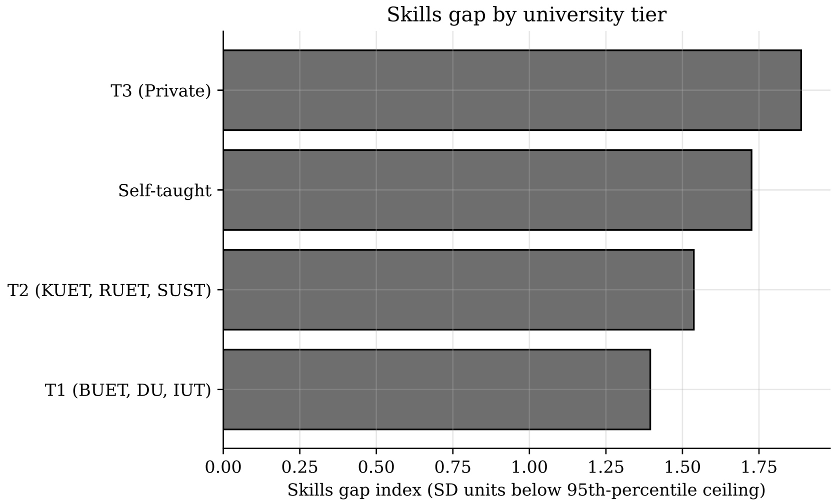

Figure 4: Skills gap index by university tier. T3 private universities, which supply the majority of the workforce, exhibit the largest remediation task.

5.6 Policy Scenarios and Workforce Projection (RQ5)

Figure 5 presents four scenarios projected for 60 months beyond May 2026. The baseline reaches 83 per cent adoption by 2031. The digital literacy scenario (raising p by 50 per cent) advances the inflection point by approximately five months. The industry mentorship scenario (raising q by 30 per cent) produces a steeper terminal trajectory. The combined scenario dominates throughout, reaching 96 per cent adoption by 2031, a 13-percentage- point gain over the baseline.

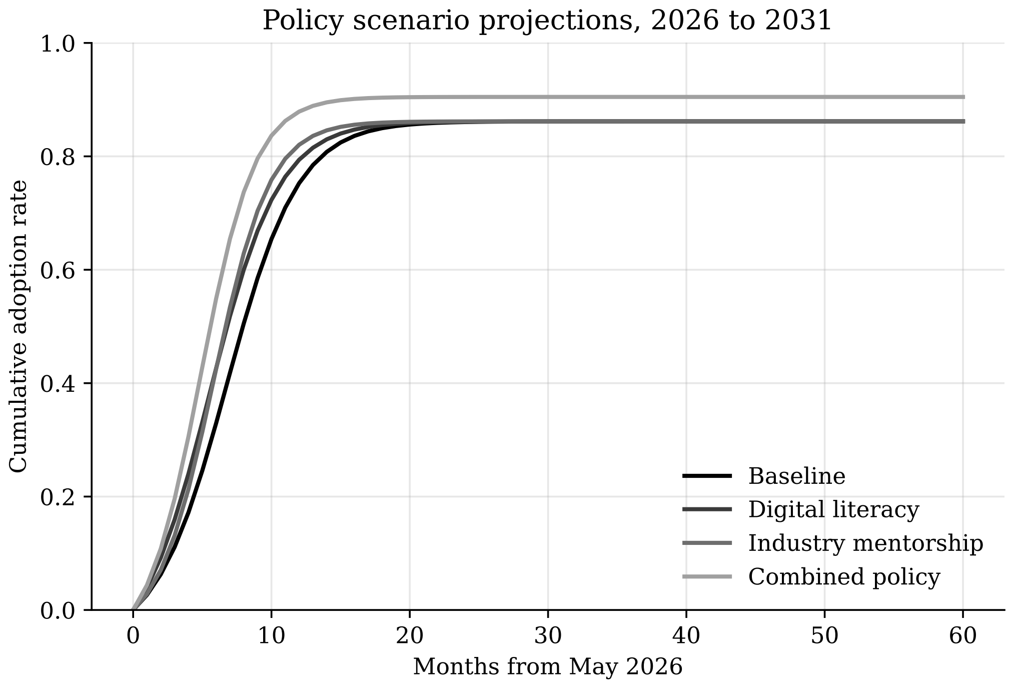

Figure 5: Policy scenario projections, May 2026 to May 2031. The combined policy reaches approximately 96 per cent adoption against a baseline of 83 per cent.

In a workforce projected to grow at 8 per cent per year (Bangladesh Association of Software and Information Services, 2025) to approximately 2.0 million practitioners by 2030, the 13-percentage-point combined-policy gain corresponds to roughly 260,000 additional adopters relative to the baseline trajectory. Figure 6 positions Bangladesh against a comparator panel of emerging economies. Bangladesh currently sits slightly below the Vietnam-Philippines trajectory implied by its per capita GDP, suggesting meaningful headroom for targeted policy intervention.

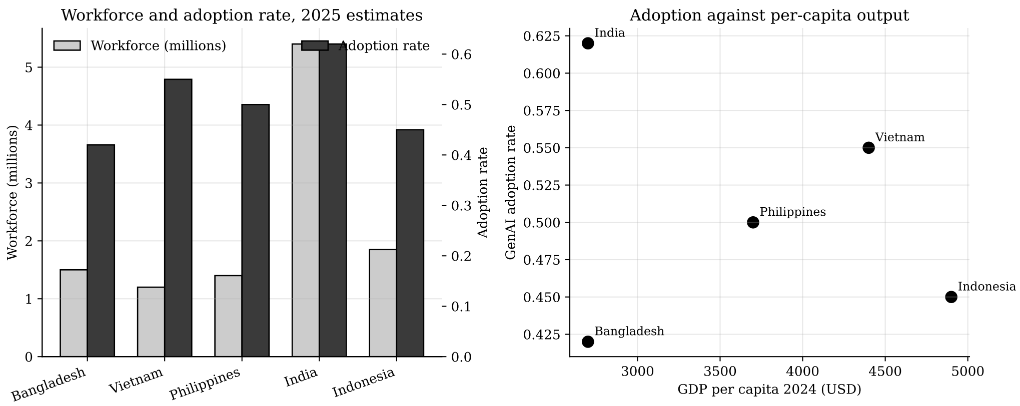

Figure 6: Workforce size and GenAI adoption rate against per capita GDP for a comparator panel. Bangladesh sits below the trajectory implied by its income level, indicating policy headroom.

6. Discussion

6.1 Adoption Premium in Context

The unadjusted Cohen's $d = 1.71$ is large by international comparison. The regression decomposition suggests approximately half reflects selection on tier, firm, and English proficiency, and the remaining half reflects a genuine treatment effect consistent with international field study evidence (Cui et al., 2025; Peng et al., 2023; Ziegler et al., 2024). The salary differential of $d = 0.28$, while smaller, is non-trivial and implies that the productivity premium is being partly appropriated by adopters, with potential implications for within-firm wage inequality.

6.2 English as a Structural Moderator

The monotone gradient from A2 to C1 is consistent with Nur and Talukder (2024) and motivates two distinct conclusions. First, the strength of the effect at A2 is encouraging for sector-wide diffusion, indicating GenAI tools are broadly accessible. Second, the size of the A2-to-C1 gradient confirms that English training programmes are a high-leverage complement to GenAI investment. This finding is specific to LMIC contexts and is absent from the dominant North American and European literature, a respect in which country-specific research supplies findings the global literature cannot.

6.3 Tier 3 Universities as the Critical Policy Node

The largest skills gap sits at the T3 tier, which also supplies the majority of the workforce. This means interventions targeted at T3 institutions carry the largest expected workforce-wide return. The international literature on GenAI in computing education (Becker et al., 2023; Denny et al., 2024; Prather et al., 2024; Vadaparty et al., 2025) identifies three immediately portable pedagogical directions: (i) scaffolded use with explicit metacognitive prompts; (ii) redesigned assessments that probe understanding rather than mere code production; and (iii) sustained attention to debugging and code review as distinctively human skills. The skills gap index (Equation 3) can serve as a quantitative baseline against which future curricular interventions are evaluated.

6.4 Policy Implications for SDG 8 and SDG 9

The combined policy scenario adds approximately 260,000 additional adopters to the workforce by 2031. This is directly relevant to SDG 8 (Decent Work) through productivity-linked wage growth, and to SDG 9 (Industry and Innovation) through the expansion of the digital capabilities of the IT export sector. The marginal cost of a national digital literacy expansion is in principle calculable from the a2i programme budget (a2i, ICT Division, Government of Bangladesh, 2024); the present results provide the productivity parameters needed to motivate a full cost-benefit calculation, which is a natural next step in partnership with a2i and the ICT Division.

6.5 Comparison with International Studies

Three points merit explicit note. First, the within-firm productivity effects are consistent in sign and order of magnitude with Ziegler et al. (2024) and Cui et al. (2025), although the unadjusted Bangladeshi differential is larger because the workforce is more heterogeneous and selection effects are stronger. Second, the heterogeneity by skill level documented by Brynjolfsson et al. (2025) is mirrored in the heterogeneity by firm type reported here. Third, the role of English as a moderator is specific to LMIC contexts: a country-specific study is required to supply this finding.

7. Limitations

Simulation, not survey. The analysis rests on a calibrated simulation rather than a primary survey. Marginal distributions are anchored to publicly reported figures, but the joint distribution is model-generated. Findings should be read as conditional on model assumptions. The full source code release is intended to allow future researchers to substitute primary survey data once available.

Causal identification. The regression decomposition controls for observable confounders but does not establish causality. A randomised intervention or a difference-in-differences design exploiting staggered Copilot for Business roll-outs within Bangladeshi MNC subsidiaries would be required to identify a causal treatment effect.

Bass model assumptions. The Bass model assumes a homogeneous market and a fixed market potential. In practice m is plausibly endogenous to the policy environment. The reported parameters should be read as a parsimonious aggregate description.

Salary modelling. The lognormal salary specification does not capture the heavy upper tail of MNC compensation. Sensitivity to alternative salary specifications is left to future work.

Generalisability. The analysis is calibrated to Bangladesh. Although several patterns are likely to generalise to comparable LMICs, the policy scenarios and workforce projection are not directly portable without country-specific recalibration.

8. Conclusion

This paper has presented the first integrated quantitative account of GenAI adoption in the Bangladeshi software engineering workforce between 2022 and 2026. The Bass diffusion parameters ($\hat{p} = 0.026$, $\hat{q} = 0.36$, $\hat{m} = 0.86$) indicate steep imitation-driven dynamics comparable to India and Vietnam but substantially faster than the least-developed-country average. A 21-percentage-point firm-type adoption gap between MNC subsidiaries and local SMEs is attributed to differences in English exposure, training infrastructure, and organisational risk tolerance. English proficiency is a robust moderator, with treatment effects rising monotonically from CEFR

A2 to C1. Skills gaps are largest at T3 private universities, the tier supplying the majority of the workforce. A combined digital literacy and industry mentorship policy is projected to add approximately 260,000 additional adopters by 2031, with direct implications for SDG 8 and SDG 9 targets.

Three lines of immediate extension are: (i) a primary survey of 2,000–5,000 practitioners in partnership with BASIS and a2i; (ii) a difference-in-differences design exploiting staggered Copilot for Business roll-outs to identify causal treatment effects; and (iii) a longitudinal pedagogical study at one or more T3 institutions to evaluate targeted curricular interventions. The Bangladeshi software sector stands at an unusually informative point in its development: large enough to matter for global software supply, heterogeneous enough to admit rigorous within-country comparative analysis, and exposed enough to GenAI tools that the diffusion is observable in real time. Policy choices made in the next two years, at the level of T3 curricula and national English and digital literacy programmes, will materially shape the Bangladeshi software workforce of 2030.

Declarations

Data and Code Availability

All source code, 40 automated unit tests, 12 figures, and 11 data tables are released at `https://github.com/tanzimkhan24/genai-se-bangladesh` under the MIT licence for code and CC BY 4.0 for the paper. The fixed random seed is `SEED=20260524`.

Conflict of Interest

The author declares no competing interests.

Ethics Statement

This study uses simulated data calibrated to publicly reported aggregate statistics. No human subjects research was conducted and no ethics approval was required.

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